Chpater1 Mechanics & Fluids

Lesson2 Dynamics: Newton's Laws of Motion

1. A woman weighs $120\ \text{lb}$. Determine **(a)** her weight in newtons and **(b)** her mass in kilograms.
2. The heaviest invertebrate is the giant squid, which is estimated to have a weight of about $2\ \text{tons}$ spread out over its length of $70\ \text{feet}$. What is its weight in newtons?
3. A football punter accelerates a football from rest to a speed of $10\ \text{m/s}$ during the time in which his toe is in contact with the ball (about $0.20\ \text{s}$). If the football has a mass of $0.50\ \text{kg}$, what average force does the punter exert on the ball?
4. A 6.0-kg object undergoes an acceleration of $2.0\ \text{m/s}^2$. **(a)** What is the magnitude of the resultant force acting on it? **(b)** If this same force is applied to a 4.0-kg object, what acceleration is produced?
5. A freight train has a mass of $1.5 \times 10^7\ \text{kg}$. If the locomotive can exert a constant pull of $7.5 \times 10^5\ \text{N}$, how long does it take to increase the speed of the train from rest to $80\ \text{km/h}$?
6. A 75-kg man standing on a scale in an elevator notes that as the elevator rises, the scale reads $825\ \text{N}$. What is the acceleration of the elevator?
7. A 970-kg car starts from rest on a horizontal roadway and accelerates eastward for $5.00\ \text{s}$ when it reaches a speed of $25.0\ \text{m/s}$. What is the average force exerted on the car during this time?
8. After falling from rest from a height of $30\ \text{m}$, a 0.50-kg ball rebounds upward, reaching a height of $20\ \text{m}$. If the contact between ball and ground lasted $2.0\ \text{ms}$, what average force was exerted on the ball?
9. If a man weighs $900\ \text{N}$ on the Earth, what would he weigh on Jupiter, where the free-fall acceleration is $25.9\ \text{m/s}^2$?
10. A 3.00-kg object undergoes an acceleration given by $\vec{a} = (2.00\vec{i} + 5.00\vec{j})\ \text{m/s}^2$. Find **(a)** the resultant force acting on the object and **(b)** the magnitude of the resultant force.
11. An airboat with mass $3.50 \times 10^2\ \text{kg}$, including the passenger, has an engine that produces a net horizontal force of $7.70 \times 10^2\ \text{N}$, after accounting for forces of resistance (see Fig. 4.6). **(a)** Find the acceleration of the airboat. **(b)** Starting from rest, how long does it take the airboat to reach a speed of $12.0\ \text{m/s}$? **(c)** After reaching that speed, the pilot turns off the engine and drifts to a stop over a distance of $50.0\ \text{m}$. Find the resistance force, assuming it's constant.

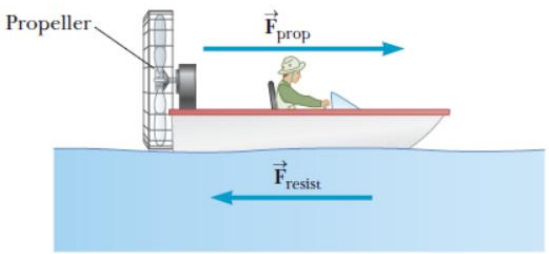


Figure 4.6 (Example 4.1)

12. Two horses are pulling a barge with mass 2.00×10^3 kg along a canal, as shown in **Figure 4.7**. The cable connected to the first horse makes an angle of $\theta_1 = 30.0°$ with respect to the direction of the canal, while the cable connected to the second horse makes an angle of $\theta_2 = -45.0°$. Find the initial acceleration of the barge, starting at rest, if each horse exerts a force of magnitude 6.00×10^2 N on the barge. Ignore forces of resistance on the barge.

13. A traffic light weighing 1.00×10^2 N hangs from a vertical cable tied to two other cables that are fastened to a support, as in **Figure 4.14a.** The upper cables make angles of $37.0°$ and $53.0°$ with the horizontal. Find the tension in each of the three cables.

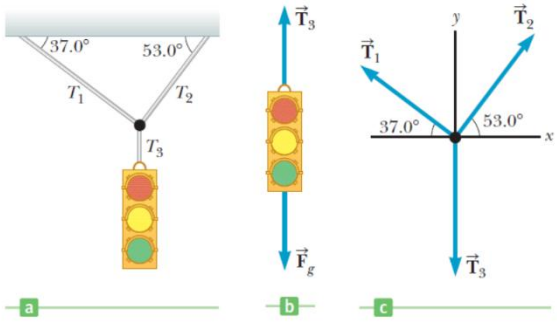


Figure 4.14 (Example 4.6) (a) A traffic light suspended by cables. (b) The forces acting on the traffic light. (c) A free-body diagram for the knot joining the cables.

14. The combined weight of the crate and dolly in **Figure 4.17** is 3.00×10^2 N. If the man pulls on the rope with a constant force of 20.0 N, what is the acceleration of the system (crate plus dolly), and how far will it move in 2.00 s? Assume the system starts from rest and that there are no friction forces opposing the motion.

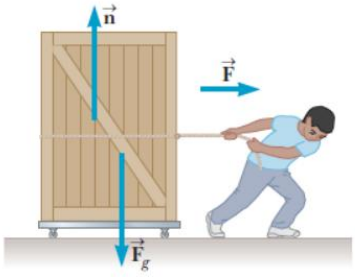


Figure 4.17 (Example 4.8)

15. A man pulls a 50.0-kg box horizontally from rest while exerting a constant horizontal force, displacing the box 3.00 m in 2.00 s. Find the force the man exerts on the box. (Ignore friction.)

16. (a) A car of mass m is on an icy driveway inclined at an angle $\theta_1 = 20.0°$, as in **Figure 4.18a**. Determine the acceleration of the car, assuming the incline is frictionless. **(b)** If the length of the driveway is 25.0 m and the car starts from rest at the top, how long does it take to travel to the bottom? **(c)** What is the car's speed at the bottom?

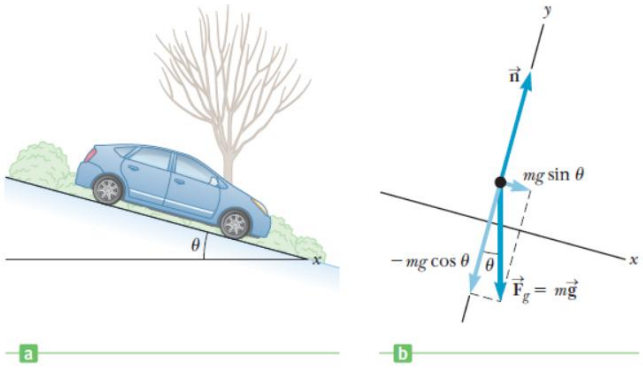


Figure 4.18 (Example 4.9)

17. **(a)** Suppose a hockey puck slides down a frictionless ramp with an acceleration of $5.00\ \text{m/s}^2$. What angle does the ramp make with respect to the horizontal? **(b)** If the ramp has a length of $6.00\ \text{m}$, how long does it take the puck to reach the bottom? **(c)** Now suppose the mass of the puck is doubled. What's the puck's new acceleration down the ramp? ANSWER **(a)** $30.7°$ **(b)** $1.55\ \text{s}$ **(c)** unchanged, $5.00\ \text{m/s}^2$

18. A woman weighs a fish with a spring scale attached to the ceiling of an elevator, as shown in **Figures 4.19a** and **4.19b.** While the elevator is at rest, she measures a weight of $40.0\ \text{N}$. **(a)** What weight does the scale read if the elevator accelerates upward at $2.00\ \text{m/s}^2$? **(b)** What does the scale read if the elevator accelerates downward at $2.00\ \text{m/s}^2$, as in **Figure 4.19b**? **(c)** If the elevator cable breaks, what does the scale read?

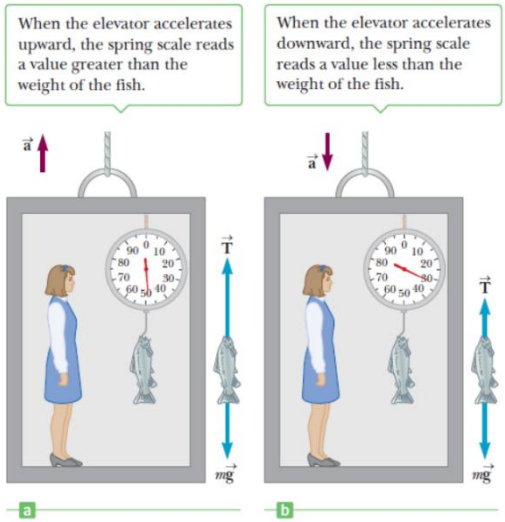
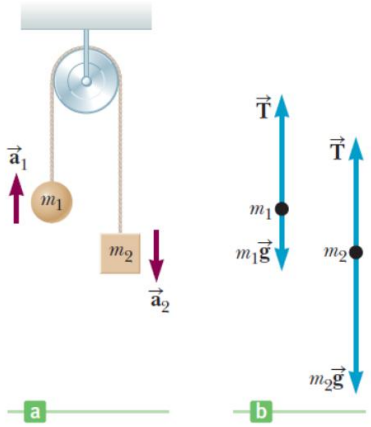


Figure 4.19 (Example 4.10)

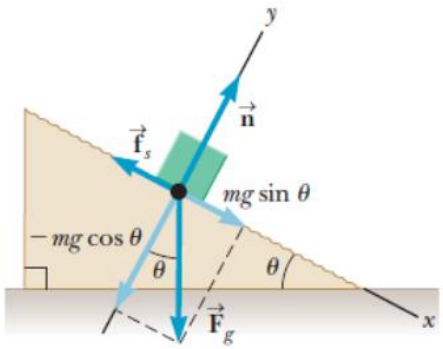
(*Continued*)



Active Figure 4.20
(Example 4.11) Atwood's machine. (a) Two hanging objects connected by a light string that passes over a frictionless pulley. (b) Free-body diagrams for the objects.

19. Two objects of mass m_1 and m_2, with $m_2 > m_1$, are connected by a light, inextensible cord and hung over a frictionless pulley, as in Active **Figure 4.20a**. Both cord and pulley have negligible mass. Find the magnitude of the acceleration of the system and the tension in the cord.

20. Suppose a block with a mass of $2.50\ \text{kg}$ is resting on a ramp. If the coefficient of static friction between the block and ramp is 0.350, what maximum angle can the ramp make with the horizontal before the block starts to slip down?



Active Figure 4.23 (Example 4.12)

21. The ramp in **Example 4.20** is roughed up and the experiment repeated. **(a)** What is the new coefficient of static friction if the maximum angle turns out to be $30.0°$? **(b)** Find the maximum static friction force that acts on the block. ANSWER **(a)** 0.577 **(b)** $12.3\ \text{N}$

22. The hockey puck in **Figure 4.24**, struck by a hockey stick, is given an initial speed of $20.0\ \text{m/s}$ on a frozen pond. The puck remains on the ice and slides $1.20 \times 10^2\ \text{m}$, slowing down steadily until it comes to rest. Determine the coefficient of kinetic friction between the puck and the ice.

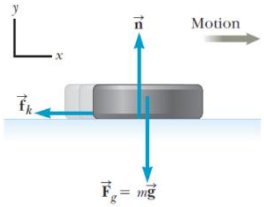


Figure 4.24 (Example 4.13)

23. (a) A block with mass $m_1 = 4.00\ \text{kg}$ and a ball with mass $m_2 = 7.00\ \text{kg}$ are connected by a light string that passes over a frictionless pulley, as shown in **Figure 4.25a.** The coefficient of kinetic friction between the block and the surface is 0.300. Find the acceleration of the two objects and the tension in the string. **(b)** Check the answer for the acceleration by using the system approach.

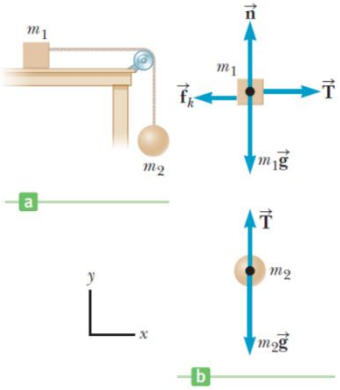


Figure 4.25 (Example 4.14) (a) Two objects connected by a light string that passes over a frictionless pulley. (b) Force diagrams for the objects.

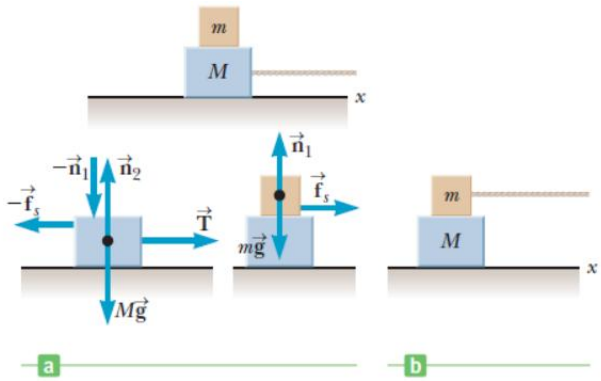


Figure 4.26 (a) (Example 4.15) (b) (Exercise 4.15)

24. A block of mass $m = 5.00$ kg rides on top of a second block of mass $M = 10.0$ kg. A person attaches a string to the bottom block and pulls the system horizontally across a frictionless surface, as in **Figure 4.26a.** Friction between the two blocks keeps the 5.00-kg block from slipping off. If the coefficient of static friction is 0.350, **(a)** what maximum force can be exerted by the string on the 10.0-kg block without causing the 5.00-kg block to slip? **(b)** Use the system approach to calculate the acceleration.

25. Assume the three blocks portrayed in **Figure P4.29** move on a frictionless surface and a 42-N force acts as shown on the 3.0-kg block. Determine **(a)** the acceleration given this system, **(b)** the tension in the cord connecting the 3.0-kg and the 1.0-kg blocks, and **(c)** the force exerted by the 1.0-kg block on the 2.0-kg block.

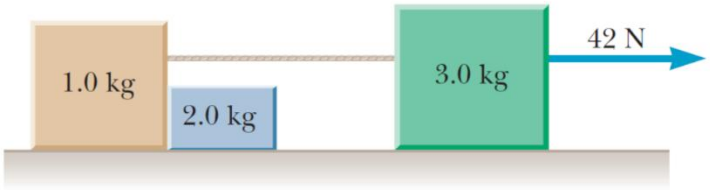


Figure P4.29

26. A block of mass $m = 5.8$ kg is pulled up a $\theta = 25°$ incline as **in Figure P4.30** with a force of magnitude $F = 32$ N. **(a)** Find the acceleration of the block if the incline is frictionless. **(b)** Find the acceleration of the block if the coefficient of kinetic friction between the block and incline is 0.10.

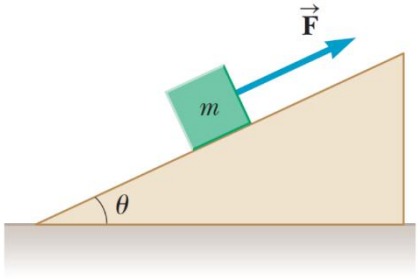


Figure P4.30

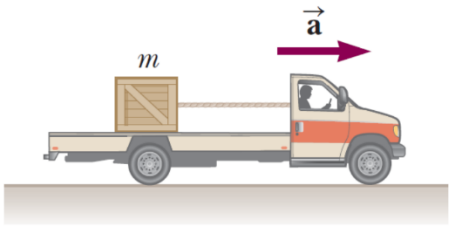


Figure P4.26

27. A crate of mass $m = 32$ kg rides on the bed of a truck attached by a cord to the back of the cab as in **Figure P4.26.** The cord can withstand a maximum tension of 68 N before breaking. Neglecting friction between the crate and truck bed, find the maximum acceleration the truck can have before the cord breaks.

28. Two blocks of masses m and $2m$ are held in equilibrium on a frictionless incline as in **Figure P4.27.** In terms of m and u, find **(a)** the magnitude of the tension T_1 in the upper cord and **(b)** the magnitude of the tension T_2 in the lower cord connecting the two blocks.

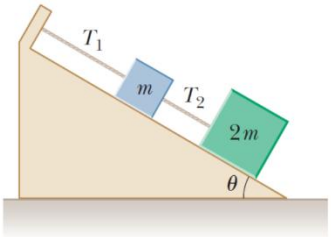


Figure P4.27

29. Two packing crates of masses 10.0 kg and 5.00 kg are connected by a light string that passes over a frictionless pulley as in **Figure P4.28**. The 5.00-kg crate lies on a smooth incline of angle 40.0°. Find **(a)** the acceleration of the 5.00-kg crate and **(b)** the tension in the string.

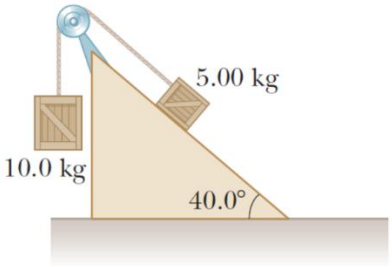


Figure P4.28

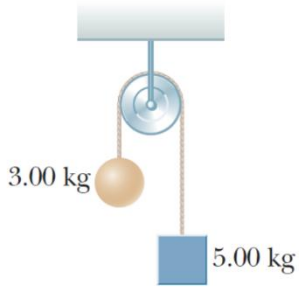


Figure P4.38

30. Two objects with masses of 3.00 kg and 5.00 kg are connected by a light string that passes over a frictionless pulley, as in **Figure P4.38**. Determine **(a)** the tension in the string, **(b)** the acceleration of each object, and **(c)** the distance each object will move in the first second of motion if both objects start from rest.

31. A dockworker loading crates on a ship finds that a 20-kg crate, initially at rest on a horizontal surface, requires a 75-N horizontal force to set it in motion. However, after the crate is in motion, a horizontal force of 60 N is required to keep it moving with a constant speed. Find the coefficients of static and kinetic friction between crate and floor.

32. In **Figure P4.36**, $m_1 = 10$ kg and $m_2 = 4.0$ kg. The coefficient of static friction between m1 and the horizontal surface is 0.50, and the coefficient of kinetic friction is 0.30. **(a)** If the system is released from rest, what will its acceleration be? **(b)** If the system is set in motion with m2 moving downward, what will be the acceleration of the system?

33. A 1 000-N crate is being pushed across a level floor at a constant speed by a force $\vec{F}$ of 300 N at an angle of 20.0° below the horizontal, as shown in **Figure P4.41a. (a)** What is the coefficient of kinetic friction between the crate and the floor? **(b)** If the 300-N force is instead pulling the block at an angle of 20.0° above the horizontal, as shown in **Figure P4.41b,** what will be the acceleration of the crate? Assume that the coefficient of friction is the same as that found in part **(a).**

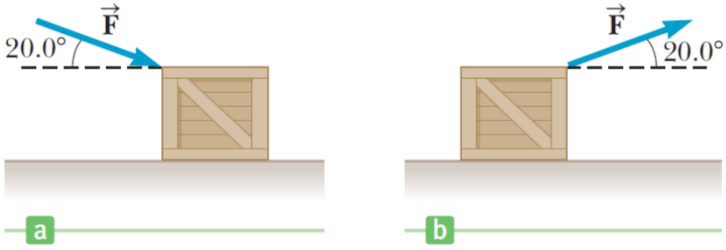


Figure P4.41